Jason Bak

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EXPERIENCE

HubSpot | *Senior Software Engineer* 2020 - 2025
Infra Costs Seattle, WA
*Worked within Platform Infrastructure, building from scratch AWS cost optimization and efficiency systems. Achieved **significant cost savings of $5M** at a company with **$210M** in annual cloud spend.*

- Identified waste patterns, advocated for prioritization, and implemented an automated resource optimization system across high-cost resource types, **eliminating $2M in accrued waste** with **zero operational disruptions** spanning all organizations.
- Built a multi-layered automated enterprise cost allocation tagging system (event-driven workers, scheduled remediation jobs, creation-time injection), achieving **$208M annual cost coverage**. This is the foundation of our cost allocation datasets, alerting infrastructure, and optimization tools.
- Architected configurable cost anomaly alerting system with custom grouping dimensions, multiple threshold types (percentage, scalar, statistical), alert muting, and team-based routing, **preventing $1M in runaway costs** and **mentoring teammates** on implementation.
- Built a centralized AWS resource metadata service, **tracking 2M resource states**, enabling historical state queries for cost data reprocessing and simplifying data retrieval across internal sources and AWS.
- **Improved annual cost attribution accuracy by $56M** through work on a Spark ETL data pipeline that processed AWS billing records. Changes include CPU/GPU cost attribution for shared Kubernetes workloads, proportional EC2 storage cost calculations, and EBS event processors.
- Collaborated directly with infrastructure and product teams to resolve their cost spend issues (waste, inefficiency, runaway costs, "unexplained" costs), **reducing costs by $2M**.

Intuit | *Software Engineer Intern* 2019
Data Platforms Mountain View, CA

- Automated EMR cluster rotation workflow for an internal data infrastructure console, **reducing an 11-minute manual process to one click**, through orchestrated termination, polling, and recreation.

Amazon | *Software Development Engineer Intern* 2018
AI Algorithms Seattle, WA

- Enhanced anomaly detection by capturing temporal context in time-series data for the Random Cut Forest C++ library. Included Python and Java bindings for cross-language interoperability.

EDUCATION

Carnegie Mellon University | B.S. in Statistics and Machine Learning 2019
Minor in Computer Science Pittsburgh, PA
Teaching Assistant: Intro to Machine Learning, Data Structures (2x)
Research Assistant: Institute for Software Research (Machine Learning)

SKILLS

Languages: Java, Python, C++, C, R, JavaScript
Technologies: Spark, MySQL, Kafka, Elasticsearch, AWS (SQS, EventBridge, Athena, S3, DynamoDB), Maven, Grafana, Prometheus, Flask, Snowflake, Git